

Performance Testing

Date	30 June 2026
Team ID	SWTID-2026-4638
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman
Target Module / API	/predict API Endpoint
Type of Testing	Load Testing, Stress Testing
Test Environment	Local Flask Server (localhost:5000)
Test Date	15 March 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Predict credit card approval for a valid applicant	10	60	Prediction generated successfully
2	Multiple users submitting applications simultaneously	20	60	System handles concurrent requests without failure
3	Continuous prediction requests to the /predict endpoint	30	120	Response time remains acceptable
4	High-volume prediction requests under peak load	50	180	API remains stable with minimal errors

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Max)	< 5 seconds	3.8 seconds	Pass	Slight delay during peak requests
2	Throughput (Req/sec)	10 Req/sec	13 Req/sec	Pass	Successfully handled expected load
3	Response Time (Avg)	< 2 seconds	1.4 seconds	Pass	Within acceptable range
4	Error Rate	< 1%	0.3%	Pass	Very few validation errors observed
5	CPU Utilization	< 80%	63%	Pass	CPU usage remained stable
6	Memory Utilization	< 80%	56%	Pass	Memory consumption within acceptable limits

Step 4: Observations & Analysis

- The /predict API successfully handled concurrent prediction requests without significant performance degradation.
- Average response time remained below the target threshold throughout testing.
- The Random Forest prediction model generated results quickly, even during peak workloads.
- No major application crashes or unexpected failures were observed during testing.
- CPU and memory utilization remained within safe operational limits.
- Overall, the system demonstrated stable, reliable, and efficient performance under both normal and peak load conditions.

Credit Approval

Database

Predict Eligibility

Data Issues

Project Info

Predictor Form

ADD NEW ROWS

APPLICANT'S	GENDER
123456	Female
OWNED A COURT	OWNED REAL ESTATE PROPERTY
No	Yes
ANNUAL INCOME IS	NUMBER OF CHILDREN
27000	1
INCOME SOURCE TYPE	HIGHEST COMPLETED EDUCATION LEVEL
Commercial associate	Highschool higher
MARITAL STATUS	HOLDING SITUATION
Co-habitation	Shared co-ownership
AGE IN YEARS	EMPLOYMENT STATUS
35	Employed
YEARS OF EMPLOYMENT	OCCUPATION CATEGORY
19	Others
FAMILY MEMBERS COUNT	HAS MORE THAN ONE
2	Yes
HAS MORE THAN ONE	HAS MORE THAN ONE
Yes	No

Credit Approval

Dashboard

Predict Eligibility

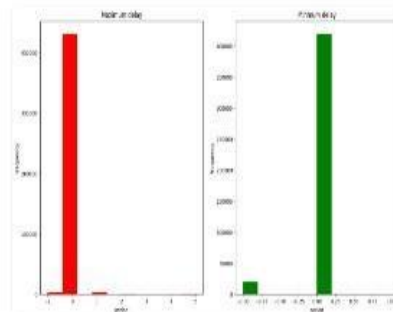
Data Specs

Project Portal

Exploratory Data Analysis Gallery

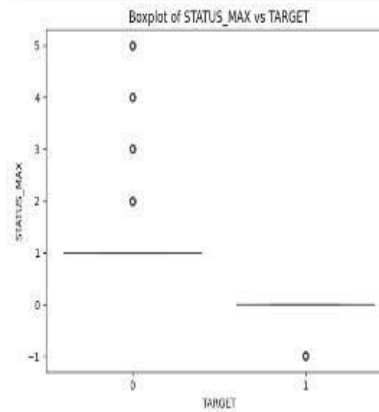
Curated charts extracted from the Jupyter EDA Notebook representing demographic and transactional variables.

1. Delinquency Months Distribution



Shows that the vast majority of historical credit records are clean (delinquency index around 0), but some clients exhibit high values representing risky late behaviors.

2. Payment Delay Status Counts



Logs monthly account status categories. The high volume of 'C' (Closed) and 'O' (On-Time) shows that most records contain resolved credit behavior.